

REINFORCEMENT LEARNING STRATEGY

Curriculum Learning in Action

Accelerating Agent Convergence Through Pedagogical Scaffolding

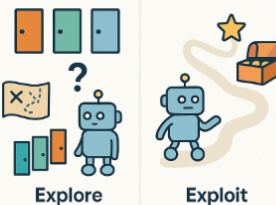
What is Curriculum Learning?

Pedagogical Scaffolding for Agents

Instead of deploying an untrained agent directly into highly complex environments with sparse rewards, **Curriculum Learning (CL)** exposes the agent to an ordered series of tasks.

- ✓ Starts with simplified tasks (e.g. closer start coordinates, less noise).
- ✓ Progresses incrementally toward the high-dimensional target task.
- ✓ Utilizes transfer learning to carry over neural network weights and policies.

Exploration vs Exploitation in Reinforcement Learning



Created by Mehul Ligade - x.com/MehulLigade

Its Place in the RL Landscape



Task Generation

Functions as a layer within environment design. It modifies transition functions, dynamically updates state spaces, or scales obstacle complexity over training steps.



Transfer Learning

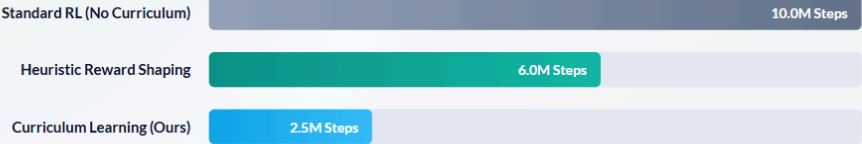
Secures the linkage between tasks. It systematically transfers value estimates, behavioral experiences, or policy networks from subtask M_{t-1} directly to subtask M_t .



Exploration Guide

Fills the void where traditional reward shaping fails. By organizing reachable target milestones, it prevents random exploration bottlenecks in sparse environments.

The Pros: Superior Efficiency



Pros: Solves catastrophic sparse reward failures, prevents local optima trapping, and lowers sample complexity by focusing exploration on reachable boundaries.

Cons and Design Pitfalls

The Handcrafting Bottleneck

Manually orchestrating task steps requires high domain intelligence. If difficulty scales too fast, agents fail to generalize. If it scales too slowly, sample inefficiencies are worsened relative to vanilla RL configurations.

Autocurriculum Stability

Automatic Curriculum Learning (ACL) generates sequences dynamically (e.g., using Teacher-Student formulations). However, this creates a meta-optimization task which is notoriously unstable and hard to converge.

Questions?

Curriculum Learning in Reinforcement Learning

 AI Research Team

Image Sources



https://miro.medium.com/0*1yXvG0ZS-6tH54W-

Source: pub.towardsai.net